

Maths Assignment 1

Q1a) $f(x) = \sqrt{4 - \sqrt{x}}$

Domain = $0 \leq x \leq 16$

for range

$$f(0) = \sqrt{4 - \sqrt{0}} = 2$$

$$f(16) = \sqrt{4 - \sqrt{16}} = 0$$

$\therefore$ Range = $0 \leq f(x) \leq 2$

b) $f(x) = \frac{2-3x}{7-2x}$

$$7-2x=0$$

$$x = \frac{7}{2}$$

$\therefore$ Domain = $x \neq \frac{7}{2}$

for range

$$y = \frac{2-3x}{7-2x}$$

$$-2xy + 7y = 2 - 3x$$

$$7y - 2 = 2xy - 3x$$

$$7y - 2 = x(2y - 3)$$

$$\frac{7y-2}{2y-3} = x$$

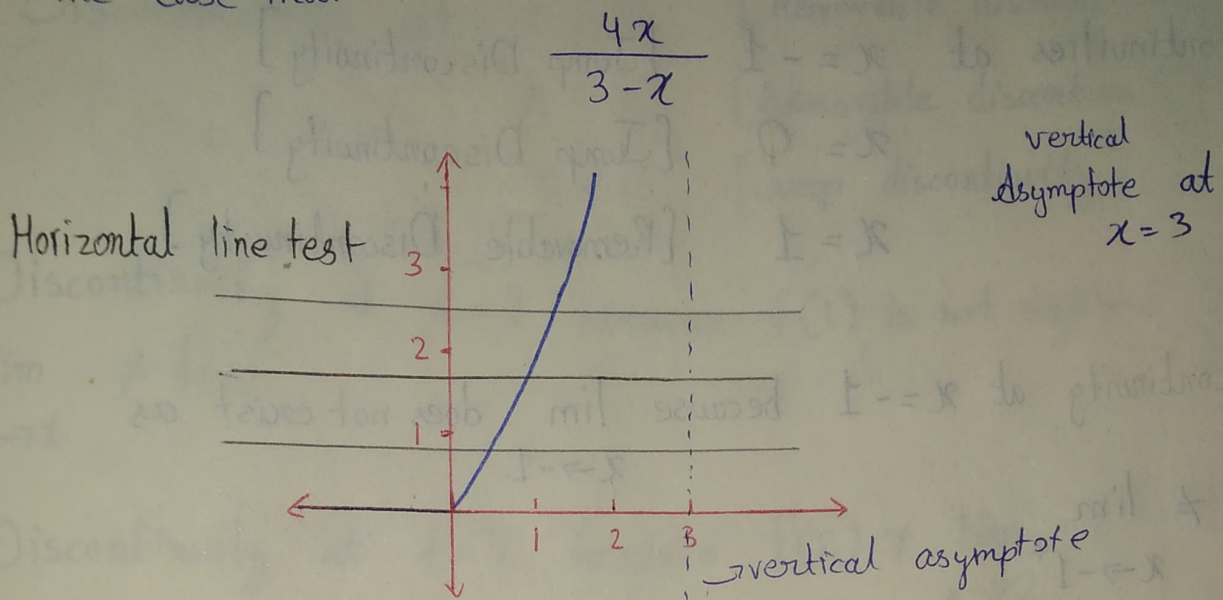
$$\therefore f^{-1}(x) = \frac{7y-2}{2y-3}$$

Domain of $f^{-1}(x) = x \neq \frac{3}{2}$

as Domain of $f^{-1}(x) =$ Range of $f(x)$

$\therefore$ Range of $f(x)$ is $f(x) \neq \frac{3}{2}$

Q2) show that the function $f: (0,1) \rightarrow (0,2)$ is one-one in the case that.



Through horizontal line test it is proven that the function is one to one as the test line only touches the graph at one point

method 2: Algebraic testing

$$f(x_1) = f(x_2)$$

$$\frac{4x_1}{3-x_1} = \frac{4x_2}{3-x_2}$$

$$4x_1(3-x_2) = 4x_2(3-x_1)$$

$$12x_1 - 4x_1x_2 = 12x_2 - 4x_1x_2$$

$$12x_1 = 12x_2$$

$$x_1 = x_2$$

hence proved

Figure 1

Discontinuities at $x = -1$ [Jump Discontinuity]

$x = 0$ [Jump Discontinuity]

$x = 1$ [Removable Discontinuity]

Discontinuity at $x = -1$ because $\lim_{x \rightarrow -1}$ does not exist as

$$\lim_{x \rightarrow -1^-} \neq \lim_{x \rightarrow -1^+}$$

Discontinuity at $x = 0$ because $\lim_{x \rightarrow 0}$ does not exist as

$$\lim_{x \rightarrow 0^-} \neq \lim_{x \rightarrow 0^+}$$

Discontinuity at $x = 1$ because $\lim_{x \rightarrow 1} \neq f(1)$ because $f(1)$ is not defined.

Graph is continuous in the following intervals

~~$(-\infty, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \infty)$~~

$$(-\infty, -1] \cup (-1, 0) \cup (0, 1) \cup (1, \infty)$$

Q3a) Figure 2

Discontinuities at $x = 1$ [Removable discontinuity]
 $x = 2$ [Removable discontinuity]
 $x = 3$ [Jump discontinuity]

b) Discontinuity at $x = 1$ because $f(1)$ is not defined hence
 $\lim_{x \rightarrow 1} \neq f(1)$

Discontinuity at $x = 2$ because $f(2) \neq \lim_{x \rightarrow 2}$

Discontinuity at $x = 3$ because $\lim_{x \rightarrow 3^+} \neq \lim_{x \rightarrow 3^-}$ hence $\lim_{x \rightarrow 3}$ does not exist

c) Graph is continuous in the following intervals,
 $[0, 1) \cup (1, 2) \cup (2, 3] \cup (3, 4]$

$$\left. \begin{array}{l} 0 \leq x < 1 \\ 1 \leq x < 2 \\ 2 \leq x < 3 \\ 3 \leq x < 4 \\ 4 \leq x \end{array} \right\} f(x) = \begin{cases} x-1 \\ x \\ x+1 \\ 0 \\ 1 \end{cases}$$

Q3a) Figure 3

Discontinuities at $x=0$ [Jump Discontinuity]
 $x=1$ [Removable Discontinuity]
 $x=2$ [Removable Discontinuity]
 $x=3$ [Removable Discontinuity]

b) Discontinuity at $x=0$ because $\lim_{x \rightarrow 0}$ doesn't exist as

$$\lim_{x \rightarrow 0^-} \neq \lim_{x \rightarrow 0^+}$$

Discontinuity at $x=1$ because $f(1) \neq \lim_{x \rightarrow 1}$

Discontinuity at $x=2$ and $x=3$ because $f(2)$ and $f(3)$ are not defined

c) Graph is continuous in the following intervals
 $[-1, 0) \cup (0, 1) \cup (1, 2) \cup (2, 3)$

d) Piece wise function

$$f(x) = \begin{cases} x^2 - 1 & -1 \leq x < 0 \\ 2x & 0 < x < 1 \\ -2x + 4 & 1 < x < 2 \\ 0 & 2 < x < 3 \\ 1 & x = 1 \end{cases}$$

$$Q4) \quad f(x) = \begin{cases} 4x^2 + ax + b & \text{if } x < 3 \\ a + b - 2 & \text{if } x = 3 \\ 2x^3 - bx + a & \text{if } x > 3 \end{cases}$$

$$4x^2 + ax + b = a + b - 2$$

$$4x^2 + ax - a + 2 = 0$$

substitute 3

$$4(3)^2 + a(3) - a + 2 = 0$$

$$2a + 38 = 0$$

$$a = -\frac{38}{2}$$

$$a = -19$$

$$2x^3 - bx + a = a + b - 2$$

$$2x^3 - bx - b + 2 = 0$$

substitute 3

$$2(3)^3 - b(3) - b + 2 = 0$$

$$-4b + 56 = 0$$

$$b = \frac{56}{4}$$

$$b = 14$$

$\therefore$ value of $a = -19$

value of $b = 14$

hence

$$f(x) = \begin{cases} 4x^2 - 19x + 14 & \text{if } x < 3 \\ -7 & \text{if } x = 3 \\ 2x^3 - 14x - 19 & \text{if } x > 3 \end{cases}$$

$$\text{Q5a)} \quad L = \lim_{x \rightarrow \infty} \frac{2x^4 - 4x^2 + 5}{3x^4 - 7x + 2}$$

$$L = \frac{x^4 \left(2 - \frac{4}{x^2} + \frac{5}{x^4} \right)}{x^4 \left(3 - \frac{7}{x^3} + \frac{2}{x^4} \right)}$$

$$L = \frac{2}{3} \quad \text{Answer}$$

$$\text{b)} \quad M = \lim_{x \rightarrow 3^-} \frac{x^3 - 5x + 4}{x^3 - 8x - 3}$$

$$= \frac{x^3 \left(1 - \frac{5}{x^2} + \frac{4}{x^3} \right)}{x^3 \left(1 - \frac{8}{x^2} - \frac{3}{x^3} \right)}$$

$$= \frac{1 - \frac{5}{3^2} + \frac{4}{3^3}}{1 - \frac{8}{3^2} - \frac{3}{3^3}}$$

$$= \frac{\frac{8}{27}}{0}$$

$$M = \infty \quad \text{Answer}$$

$$\text{c)} \quad M = \lim_{x \rightarrow 1} \frac{3x^3 - 7x^2 + 6x - 2}{x - 1}$$

$$= \frac{(x-1)(3x^2 - 4x + 2)}{(x-1)}$$

$$= 3(1)^2 - 4(1) + 2$$

$$= 3 - 4 + 2$$

$$M = 1 \quad \text{Answer}$$